

The Price of Thought: wiring cost and the brain–nerve cord connective in the complete *Drosophila* male CNS connectome

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19 September 2026

Abstract

The complete connectome of the adult male *Drosophila* central nervous system (neuPrint male-cns:v1.0) was represented as a directed graph of 23,073 cell types resolved by hemisphere, with 490,884 connections and each type placed at the centroid of its neurons' cell bodies. Three questions were pre-registered.

First, placement. The real layout costs 0.459 times the mean of 1000 random reassignments of the same positions (none as cheap), and 0.728 times when positions are shuffled only within brain or nerve cord. It is not a local optimum: cost-reducing swaps lower the total by at least 33.6%.

Second, the neck connective of descending and ascending neurons. Its neck-crossing connections are 7.5% of edges but 24.0% of wiring cost. Their partners are enriched for high-degree types, and connective types are over-represented among hubs (odds ratio 3.22). Rich-to-rich routes through the connective exceed a degree-preserving null by 1.103 ($p = 0.001$). That excess vanishes at a lower edge threshold, however, and a model using only distance, compartment and cell class reproduces it.

Third, value. Cutting the connective removes 0.36 times the sensory-to-motor flow capacity lost with random wiring of equal total length, but 1.83 times in the brain-to-nerve-cord direction. Across cell types, wire length does not predict flow once the number of connections is controlled for.

Two descriptive analyses accompany these tests. Attributed to the neuropils, only 6.6% of the wiring budget stays inside one region, and all 48 regions large enough to test are wired internally more cheaply than permutations of their own cell types, the antennal lobes least clearly. Over connections the budget is unevenly but not extremely spread: the longest tenth hold 30.7% of it, and a lognormal fits the upper tail better than a power law.

The fly concentrates its most expensive wiring on well-connected cell types. That wiring carries no more flow per unit length than ordinary long wiring, except from brain to nerve cord.

Introduction

This report is the third part of a three-part study of one connectome, the complete male *Drosophila* central nervous system (Berg et al. 2026). ConnectomeLens asked whether the male wiring alone carries a detectable signature of the cell types annotated as sexually dimorphic. Fault Lines asked how much of the nervous system has to be removed before sensory input no longer reaches motor output. This part asks what the wiring costs in physical length, and what the most expensive part of it, the connective between brain and nerve cord, buys in return.

Wiring economy is the idea that nervous systems place their components so as to keep the total length of their connections small. Ramón y Cajal (1909–1911) stated it as a law of conservation of space, time and material. Cherniak (1994, 1995) treated it as a component-placement problem. He found the layout of *C. elegans* ganglia and of mammalian cortical areas at or near the cheapest of the arrangements he examined.

Later work qualified the result:

- Chen, Hall and Chklovskii (2006) found the *C. elegans* layout cheaper than random, yet well above the optimum they computed for it.
- Kaiser and Hilgetag (2006) showed that rearranging components can cut wiring by about 48%, because long-range projections that shorten processing paths are kept. Ahn, Jeong and Kim (2006) reached a similar conclusion.
- Gushchin and Tang (2015) found the real *C. elegans* layout 30.6% cheaper than random, while a constrained optimization of 86 interneurons cut a further 35.1%.

Placement cheaper than random is therefore not placement that is optimal, and the two claims are tested separately here.

Rich-club organization is the tendency of highly connected nodes to be more densely interconnected than their degrees alone predict (Colizza et al. 2006). In the human connectome the rich club is costly. Its connections are long and make up a disproportionate share of total wiring, and they carry a disproportionate share of the shortest communication

paths (van den Heuvel and Sporns 2011; van den Heuvel et al. 2012). The same high-cost, high-capacity pattern has been reported in *C. elegans*: the connections of an 11-neuron rich club account for 48% of total wiring cost (Towlson et al. 2013).

The fly’s neck connective is the most obvious candidate for such a backbone. Descending neurons carry signals from brain to nerve cord and ascending neurons from nerve cord to brain. Each must span the neck, so their connections cannot share in the local economy that keeps most wiring short.

Both ideas have already been examined in the fly brain, and this study extends that work rather than starting it:

- Rivera-Alba et al. (2011) found the arrangement of neurons in a lamina cartridge cheaper than a million random rearrangements.
- Salova and Kovács (2025) shuffled neuron positions across the hemibrain and found shuffled layouts about twice as costly as the real one. They also found connection probability decaying exponentially with distance, and fitted generative models combining degree with wiring length.
- Péntek and Ercsey-Ravasz (2025) fitted an exponential distance rule model to FlyWire neuropils.
- Lin et al. (2024) found a neuron-level rich club in the FlyWire brain against a degree-preserving null. They expected ascending and descending neurons to belong to a rich club spanning the whole central nervous system, and noted that testing this required a complete CNS connectome.

Two recent network analyses of the FlyWire brain are the closest prior work:

- Grindrod, Lambiotte and Sahasrabudde (2024) studied 32,272 central-brain neurons. They found 67 Infomap modules, 21 of which carry 94% of the flow, arranged in a significant SpringRank hierarchy and spatially interleaved rather than compact.
- Sulyok, Balogh and Palla (2026) compared the real three-dimensional coordinates of 132,483 neurons with hyperbolic and Node2vec embeddings. Real coordinates predict connections with an AUC of 0.862 but support greedy routing in only 7.5% of attempts, and every embedding outperforms them.

Neither paper measures wiring cost, tests a rich club, fits a generative wiring model, or contains the nerve cord. In FlyWire, ascending and descending neurons appear only as their parts inside the brain.

The two whole-CNS connectomes have already identified the connective as a bottleneck, by other measures. Berg et al. (2026) computed maximum flow from sensory modalities to motor domains and found descending and ascending neurons with much higher flow utilization than other classes. Bates et al. (2026), in a female brain-and-nerve-cord connectome, found them with high betweenness on sensory-to-effector paths. Neither measures what the connective costs in wire, whether it favours highly connected partners, or whether removing it does more damage than removing ordinary wiring of the same length.

To our knowledge, this is the first test of wiring economy and rich-club organization on a single synapse-resolution connectome of an entire central nervous system, with the brain–nerve cord connective treated as its own class of connections. It adds four things at cell-type resolution:

1. a placement permutation test spanning brain, nerve cord and connective;
2. a degree-preserving test of whether connective routes join high-degree types on the two sides, which addresses the expectation of Lin et al. (2024) directly;
3. the loss of sensory-to-motor flow capacity when the connective is removed, against removing non-connective wiring of equal total length;
4. a test of whether a generative model built on distance, compartment and cell class reproduces these statistics.

Data and Methods

Spatially embedded cell-type graph

All data were retrieved from neuPrint (male-cns:v1.0) with `neuprint-python`; coordinates were converted at the dataset’s 8 nm voxel size, and the somata of typed neurons span $730 \times 514 \times 995 \mu\text{m}$. The 164,506 typed neurons were grouped into nodes, each a cell type on one side of the body (the soma hemisphere, or the nerve-root hemisphere for neurons without a CNS soma), since pooling a bilateral type would place it on the midline. This gives 23,073 nodes.

A node sits at the centroid of its neurons’ soma positions when at least half of them have one (22,139 nodes). Otherwise it sits at the synapse-weighted centroid of its neurons’ synapses (934 nodes, 751 of them sensory, whose cell bodies lie in the periphery). A node is brain-dominant if more than half of its synapses in CentralBrain, Optic(L), Optic(R) and VNC lie in the first three, and nerve cord–dominant otherwise: 16,052 and 7,021 nodes.

A directed edge joins two nodes when the connection supplies at least 1% of the target node's input synapses from typed neurons; self-connections are excluded. The graph has 490,884 edges. The same rule applied to types pooled across sides reproduces exactly the 11,751-type, 243,439-edge graph used in ConnectomeLens and Fault Lines.

The connective

Each cell type takes the superclass held by most of its neurons. Descending types (`descending_neuron`, 480 types) and ascending types (`ascending_neuron`, 563 types) form the connective set: 951 descending and 1,096 ascending nodes, 8.9% of all nodes. As a cross-check, 85.4% of typed neurons with synapses inside the neck connective ROI (CV) are descending or ascending neurons. Sensory ascending neurons, whose axons also pass through the neck, have no CNS soma to place and are treated as sensory, not connective.

Partners are all non-connective nodes, split by compartment into brain partners B (14,950) and nerve-cord partners V (6,076). A neck-crossing edge joins a connective node to a partner on the far side of the neck: $DN \rightarrow V$, $V \rightarrow DN$, $AN \rightarrow B$ or $B \rightarrow AN$. There are 36,943 such edges.

Wiring cost

The cost of an edge is the Euclidean distance between its endpoints' positions; the cost of an edge set is the sum over its edges, each counted once. Edge lengths have median 154.3 μm and mean 189.6 μm .

Node positions stand in for the length of real neuronal processes. As a check, skeletons were retrieved for a stratified random sample of 500 neurons: 100 descending, 100 ascending, 150 central-brain intrinsic and 150 nerve-cord intrinsic. Their total cable length was compared with the distance from soma to the centroid of the neuron's presynaptic sites.

Placement test

The primary test compared the real total unweighted cost with 1000 uniform random permutations of the position-to-node assignment, with the graph and the multiset of positions unchanged. Secondary versions were:

- the synapse-weighted cost;
- permutation only within compartments;
- synapse centroids as positions for every node;
- the brain and nerve-cord subgraphs separately.

Distance from a local optimum was estimated with 2,000,000 proposed swaps of position between two soma-placed nodes of the same compartment, each accepted if it lowered the total cost, following Kaiser and Hilgetag (2006) and Gushchin and Tang (2015). Connection probability was measured in 20 μm distance bins, and an exponential $P(d) = a \cdot \exp(-d/\lambda)$ was fitted over 0–600 μm .

Wire by neuropil

The 105 published neuropil surfaces were simplified onto a 3 μm grid and pooled into one search tree. Each cell type takes the neuropil whose surface lies nearest its position, and each connection lends half its length to the neuropil at each of its ends, so the wire attributed to the regions sums to the total. A neuropil's internal connections are those with both endpoints assigned to it.

Placement inside a region was tested with the statistic of the primary placement test restricted to that region: the summed length of its internal connections against 1000 permutations of which of its own cell types sits at which of its own positions. It is reported for neuropils with at least 12 cell types and 50 internal connections.

Concentration of the wiring budget

Connections were sorted longest first and the cumulative share of wire taken against the cumulative share of connections; the Gini coefficient is one minus twice the area under the same curve taken shortest first. It was computed over all connections and over those within the brain, within the nerve cord and across the neck.

The upper tail of the length distribution was fitted with the `powerlaw` package, which selects the lower bound minimizing the Kolmogorov-Smirnov distance above it and then fits the exponent by maximum likelihood. Lengths were rounded to the micrometer first, which cuts the candidate lower bounds from half a million to about a thousand without moving the fit by more than the rounding. The fit was weighed against a lognormal, an exponential and a truncated power law by normalized log-likelihood ratio, a positive ratio favouring the power law.

Wire was also attributed to cell classes, each connection counted for the majority superclass at each of its two ends. That majority is taken within a node, and differs a little from the connective set above, which takes it over the cell type

pooled across sides: 1,089 nodes have `ascending_neuron` as their majority superclass, against the 1,096 ascending nodes of the connective set.

Rich-club tests

A partner's richness is its total degree in the subgraph of non-connective nodes, so no connective edge contributes to it. Rich brain and rich nerve-cord partners are those at or above the 90th percentile of richness within B and within V. A route is a two-edge path through one connective node, $b \rightarrow \text{DN} \rightarrow v$ or $v \rightarrow \text{AN} \rightarrow b$. The statistic R is the number of rich-to-rich routes: the sum over connective nodes of their rich inputs times their rich outputs on the opposite side.

The four layers of connective edges ($B \rightarrow \text{DN}$, $\text{DN} \rightarrow V$, $V \rightarrow \text{AN}$, $\text{AN} \rightarrow B$) were randomized independently by degree-preserving edge swaps (10 swaps per edge, no multi-edges), 1000 times. Every partner keeps its number of connective edges in each layer and every connective node keeps its in- and out-degree; only which partner attaches to which connective node changes.

Two secondary tests were run:

- whether rich partners are over-represented among connective partners, against a null that redraws each connective node's partners uniformly;
- whether descending and ascending nodes are over-represented among nodes at or above the 90th percentile of total degree (Fisher exact test).

The rich-club coefficient of the whole graph was normalized by 100 degree-preserving randomizations.

Value tests

Sensory-to-motor value uses the definitions of Fault Lines, with the code copied unchanged:

- **Flow capacity** is the maximum number of edge-disjoint directed paths from any sensory node to any motor node, computed as a unit-capacity maximum flow between a supersource and a supersink.
- **Reachable pairs** is the number of (sensory, motor) pairs joined by a directed path.

The sensory set has 751 nodes (390 brain, 361 nerve cord) and the motor set 367 nodes (86 brain, 281 nerve cord). Descending neurons, which Fault Lines counted among motor outputs, are excluded here because they are part of what is removed.

The primary test removed the 36,943 neck-crossing edges. It compared the loss with removals of 1000 random sets of non-connective edges matched to their total length: edges were taken in random order until the running total reached that length, and every set came within 0.01% of it. Secondary comparisons were:

- sets matched on edge count instead of length;
- the longest non-connective edges up to the same total length;
- silencing every edge incident on a connective node;
- flow in each direction across the neck.

For each connective node, *price* is the summed length of its neck-crossing edges. *Value* is the loss of flow in the direction the node carries when those edges alone are removed. Price and value were correlated across descending and across ascending nodes, then again with the number of neck-crossing edges partialled out.

Generative model

Edge presence over ordered node pairs was modelled by logistic regression, fitted in scikit-learn on all 490,884 edges and 2,454,420 uniformly sampled non-edges. The intercept was corrected for the sampling ratio. Model G uses:

- distance, and the logarithm of distance;
- whether the two nodes share a compartment;
- a one-hot term for the ordered pair of superclasses.

A secondary model, G+deg, adds the logarithms of source out-degree and target in-degree. Fit is reported as McFadden pseudo- R^2 , AIC and 5-fold cross-validated ROC AUC.

Fifty synthetic graphs per model were drawn as independent Bernoulli trials, with a scalar shift solved so that the expected edge count equals the real one. Each synthetic graph was compared with the real graph on thirteen properties. A property counts as reproduced when the real value lies within the central 95% of the synthetic values.

Table 1: Placement test. Each row compares the real wiring cost with 1000 permutations of node positions.

analysis	real / permuted cost	z	permutations at or below real
soma positions, unweighted (primary)	0.459	-278.8	0 / 1000
synapse-weighted cost	0.320	-37.3	0 / 1000
permutation within compartment	0.728	-156.9	0 / 1000
synapse-centroid positions	0.277	-337.4	0 / 1000
brain subgraph only	0.679	-169.8	0 / 1000
nerve-cord subgraph only	0.741	-64.7	0 / 1000

Robustness, pre-registration and software

The placement, route and hub tests were repeated at edge thresholds of 0.5%, 2% and 5%. Every hypothesis, with its direction, statistic, null and threshold, was committed before the statistic was computed (commits 39892c5, e62c6df and 73fba33); each results file keeps that section unchanged and the scripts only append beneath it. The wire atlas and the concentration of the budget carry no registered hypothesis: both were run after the registered tests and are reported as description. Empirical one-sided p-values are $(1 + \text{null values at least as extreme}) / (1 + N)$, so the smallest attainable with 1000 draws is 0.0010. Software: Python 3.12, igraph 1.0.0, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, navis 1.12.0, powerlaw 2.0.0 and neuprint-python 0.6.3, pinned in requirements.txt; make reproduce rebuilds every result from neuPrint.

Results

Placement is far cheaper than random, and far from optimal

The real placement costs 93,095 mm of wire, 0.459 times the mean of 1000 random reassignments of the same positions ($z = -278.8$). No permutation came as low ($p = 0.0010$, the smallest attainable). Every secondary analysis points the same way:

The full permutation moves brain types into the nerve cord and is an easy null to beat. Shuffling only within compartments still leaves the real layout 27% cheaper, so the economy is not just the separation of brain and nerve cord.

Connection probability falls with distance (Spearman $\rho = -0.993$ across 51 bins with at least 1000 pairs), with a length constant of 76 μm over the first 600 μm . Within the brain and within the nerve cord the fall is not monotonic. Probability reaches a minimum at 340 μm between brain nodes and at 500 μm between nerve-cord nodes. It then rises again at the longest distances, where most edges join opposite sides of the body.

The real placement is not a local optimum. Of 2,000,000 proposed swaps within a compartment, 50,050 lowered the cost. Together they reduced it from 93,095 mm to 61,778 mm, a 33.64% reduction.

The search had not converged (the last 100,000 proposals still saved 0.33%), so this is a lower bound, and it ignores every physical constraint on where cell bodies and neuropils can lie.

The cable-length check supports node positions as a between-class measure only. Across the 500 sampled neurons, skeleton cable length correlates with soma-to-output distance ($\rho = 0.428$, $p = 5.5 \times 10^{-24}$): descending and ascending neurons have more cable (medians 4,592 and 3,898 μm) than intrinsic neurons (1,428 and 1,713 μm). Within each superclass, however, $|\rho|$ is below 0.12.

Inside the neuropils, placement is economical too

Each cell type was assigned to the nearest of the 105 published neuropil surfaces. Cell types fall in 89 of them, at a median distance of 9.02 μm from the surface they are assigned to, and each connection lends half its length to the neuropil at each of its ends. Only 6.6% of the wire, 6,154 mm over 86,881 connections, stays inside one neuropil; the rest runs between them. The heaviest route between two regions joins the gnathal ganglia (GNG) to the abdominal neuromere (ANm), 1,586 mm over 2,101 connections.

The budget is not spread evenly over the regions. GNG holds 10.1% of all wire, the largest share of any neuropil, ahead of ANm at 5.8%; the six leg neuropils of the nerve cord hold between 3.1% and 4.0% each.

Of the 89 neuropils, 48 hold enough cell types and internal connections to take that permutation test. All 48 come out below their permuted mean, and in 41 of them no permutation of the region's own cell types over its own positions was as short. The most economical are the mushroom body lobe gL(L) at 0.428 of its permuted mean ($z = -10.40$) and the

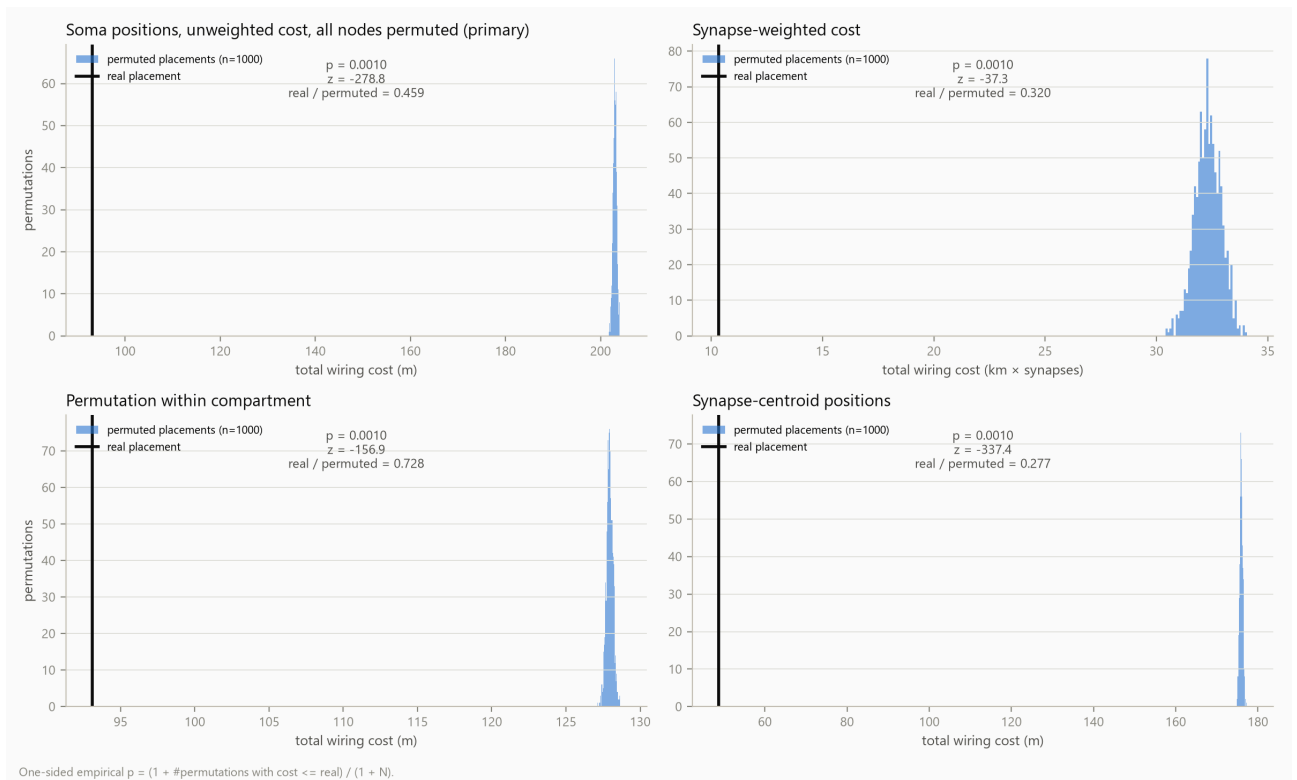


Figure 1: Real wiring cost against 1000 permutations of node positions, for the primary and three secondary analyses.

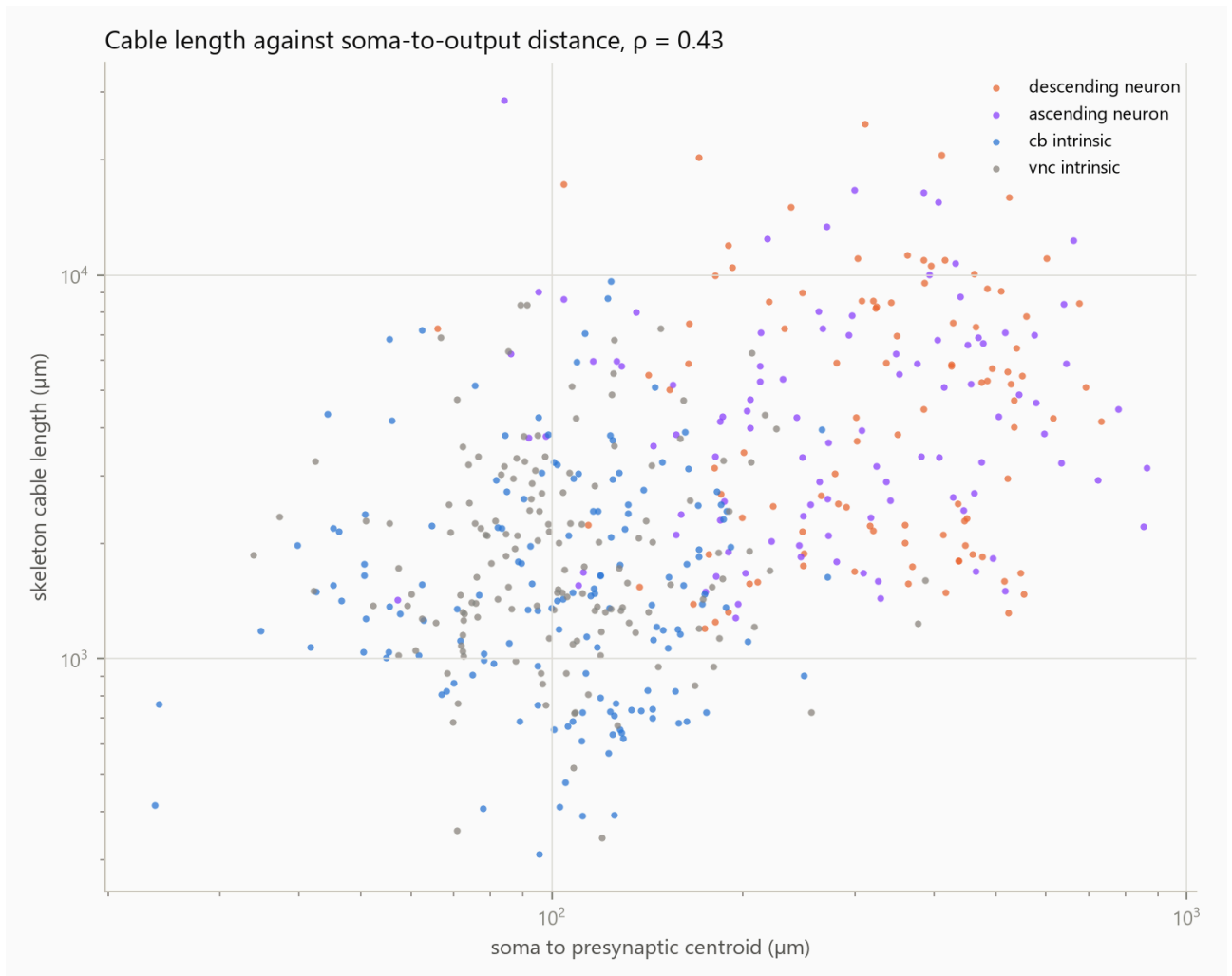


Figure 2: Skeleton cable length against soma-to-output distance for 500 sampled neurons, by superclass.

Table 2: Wire by neuropil. The twelve neuropils holding most wire, of the 89 that hold cell types. internal / permuted is the summed length of a neuropil's internal connections against the mean of 1000 permutations of which of its own cell types sits at which of its own positions.

neuropil	compartment	cell types	wire (mm)	share of wire	internal connections	internal / permuted	z
GNG	brain	1,918	9,445	10.1%	15,310	0.809	-33.39
ANm	nerve cord	1,108	5,420	5.8%	11,979	0.729	-23.94
AVLP(R)	brain	1,135	3,772	4.1%	5,881	0.863	-15.62
LegNp(T2)(R)	nerve cord	803	3,682	4.0%	3,764	0.885	-9.19
LegNp(T3)(R)	nerve cord	759	3,673	3.9%	2,603	0.894	-8.75
LegNp(T2)(L)	nerve cord	799	3,654	3.9%	3,791	0.895	-10.44
AVLP(L)	brain	1,075	3,572	3.8%	5,096	0.879	-11.18
LegNp(T1)(L)	nerve cord	667	3,192	3.4%	1,751	0.891	-7.72
LegNp(T3)(L)	nerve cord	636	3,082	3.3%	1,740	0.922	-5.72
SAD	brain	677	3,030	3.3%	1,842	0.682	-18.80
LegNp(T1)(R)	nerve cord	590	2,845	3.1%	1,493	0.885	-7.40
PB	brain	820	2,719	2.9%	2,277	0.767	-13.35

two anterior optic tubercles, AOTU(R) at 0.516 and AOTU(L) at 0.536. The economy of the layout is therefore not only the separation of brain from nerve cord, nor of one neuropil from the next: it holds again inside nearly every region, over the positions that region already has.

Three regions are the exception, and not for want of data. The antennal lobes come closest to chance: AL(L) is at 0.974 of its permuted mean with 171 of 1000 permutations as short or shorter ($z = -1.00$), and AL(R) at 0.960 with 53 ($z = -1.62$). The accessory medulla AME(R) is at 0.939 with 231 ($z = -0.75$). AL(L) holds 551 cell types and 3,139 internal connections, more than all but a few of the regions tested, and the two hemispheres agree. Inside the antennal lobes, the arrangement of cell types over their own positions saves little or no wire.

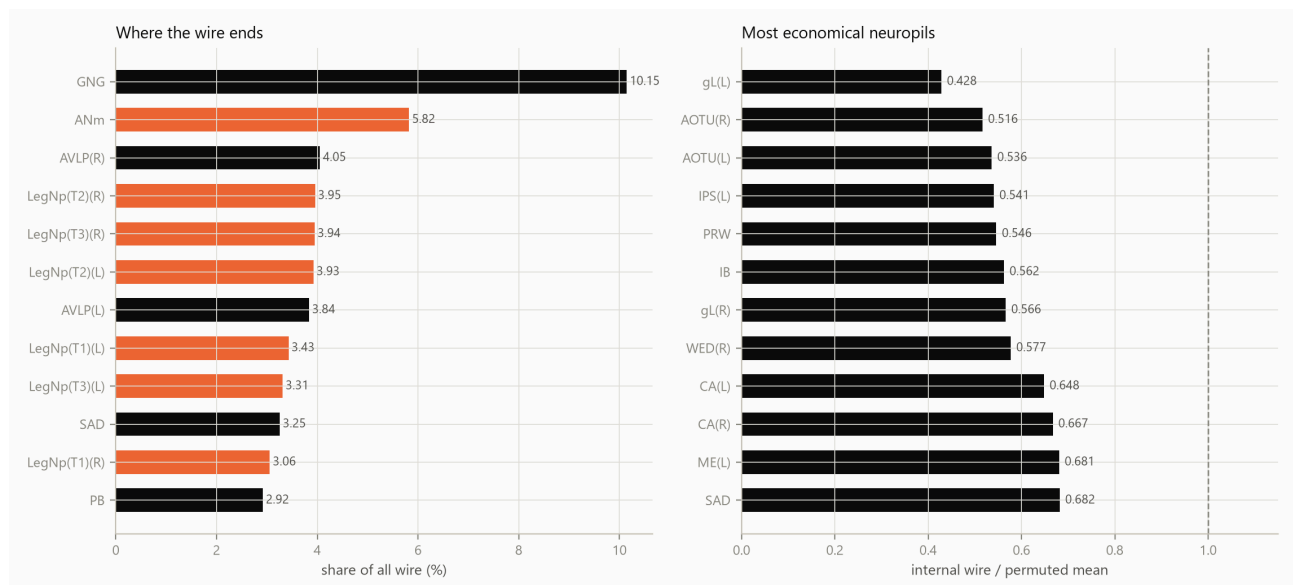


Figure 3: The wire held by the twelve neuropils that hold most of it, and how economically each is wired internally.

Table 3: Length and concentration by where a connection runs. The groups are views of the same 490,884 connections, not a partition: a connection between a brain type and a nerve-cord type that is neither descending nor ascending falls in none of the last three.

group	connections	wire (mm)	mean (μm)	median (μm)	99th percentile (μm)	Gini	longest 10%
all	490,884	93,095	189.7	154.3	808.8	0.404	30.7%
within the brain	319,431	49,540	155.1	146.3	544.7	0.343	23.6%
within the nerve cord	133,277	21,855	164.0	145.2	570.5	0.334	24.0%
across the neck	36,943	22,371	605.6	597.6	941.4	0.151	14.6%

The wiring budget is concentrated, and its upper tail is not a power law

The 490,884 connections hold 93,095 mm of wire between them, and they hold it unequally. Sorted longest first, the longest 1% hold 4.6% of the budget, the longest 10% hold 30.7% and the longest quarter 52.2%; the Gini coefficient of the length distribution is 0.404. That is uneven, but a long way from the concentration the rich-club literature deals with, where an eleven-neuron club holds 48% of the wiring cost of *C. elegans* (Towlson et al. 2013).

Neck-crossing connections are the longest and also the most uniform. Their median length, 597.6 μm , is four times the 146.3 μm within the brain, while their Gini, 0.151, is less than half the 0.343 within it. Crossing the neck costs about the same whichever pair of cell types does it.

The upper tail is heavier than an exponential, but a power law is not the best description of it. Fitted above 161 μm , over 231,054 connections, the exponent is 2.99. By normalized log-likelihood ratio the power law beats an exponential (46.81) and loses to both a lognormal (-45.52) and a truncated power law (-62.52), each comparison with $p < 0.0001$. The longest connection in the graph is 1,003.9 μm , against the 995 μm the somata span along their longest axis: the distribution is cut off by the extent of the nervous system itself, which is what the truncated fit picks up and the plain power law cannot.

Central brain intrinsic types, 13,027 of the 23,073 nodes, touch connections holding 55.7% of the wire, and nerve cord intrinsic types 32.0%. The 1,089 ascending and 951 descending types touch 22.5% and 20.4% of it, and their connections

Table 4: Wire by cell class, for the eight classes that own most of it. A connection is counted for the class at each of its two ends, so the shares add to more than one.

class	cell types	connections	wire (mm)	share of wire	mean length (μm)	median (μm)
central brain intrinsic	13,027	300,466	51,887	55.7%	172.7	152.0
nerve cord intrinsic	5,393	141,088	29,829	32.0%	211.4	158.1
ascending	1,089	57,516	20,906	22.5%	363.5	306.7
descending	951	49,647	19,013	20.4%	383.0	332.9
visual projection	681	31,050	4,866	5.2%	156.7	149.8
nerve cord sensory	328	18,765	2,730	2.9%	145.5	123.9
optic lobe intrinsic	529	17,415	2,338	2.5%	134.2	127.0
central brain sensory	342	13,299	1,574	1.7%	118.4	97.1

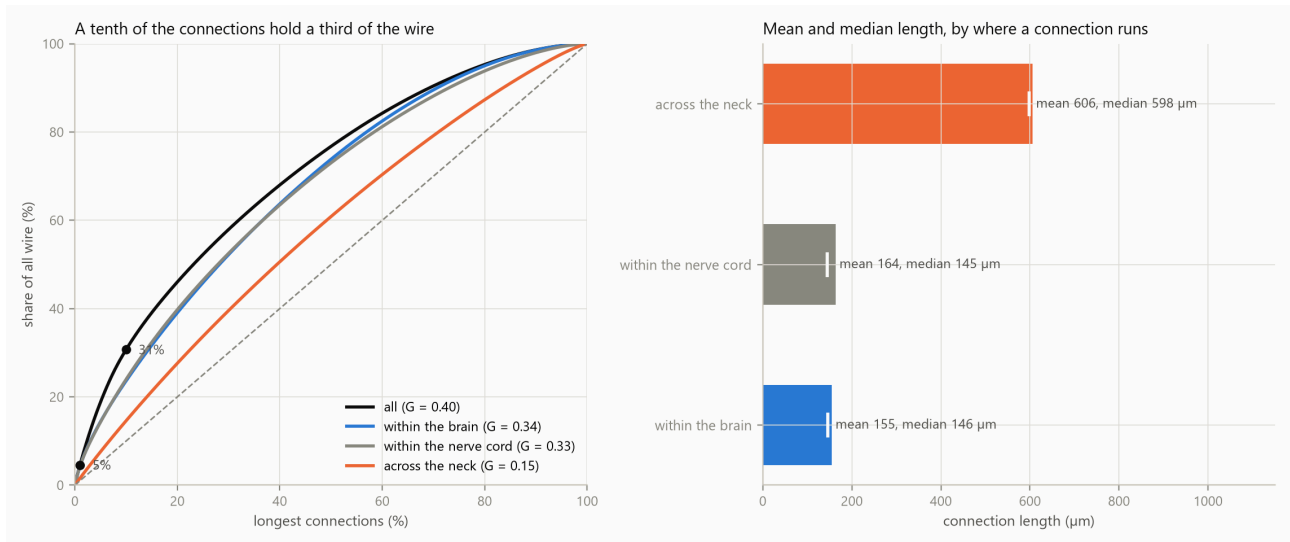


Figure 4: The Lorenz curve of the wiring budget, and the length of the connections behind it.

are the longest of any class with more than a thousand connections: mean 363.5 and 383.0 μm against 172.7 μm for central brain intrinsic types. What that expense buys is the subject of the sections that follow.

The connective is expensive, and its partners are well connected

Neck-crossing edges are 7.5% of all edges and 24.0% of total wiring cost, with a mean length of 606 μm against 189.6 μm for all edges. Counting every edge incident on a connective node, the connective carries 20.4% of edges and 38.7% of cost.

Edges incident on high-degree nodes (total degree at least 67, the 90th percentile) are longer than other edges: mean 211 against 175 μm (one-sided Mann-Whitney $p < 10^{-300}$). The high-cost backbone reported in human and *C. elegans* connectomes is therefore present here. Long edges also carry somewhat more traffic: edge length correlates with directed edge betweenness at $\rho = 0.121$, and at $\rho = 0.084$ among edges within one compartment.

Descending and ascending nodes are over-represented among the most connected nodes of the whole graph. At or above the 90th percentile of total degree lie 24.4% of descending nodes, 23.2% of ascending nodes and 8.8% of other nodes (odds ratio 3.22, Fisher $p = 1.5 \times 10^{-79}$). This is the expectation of Lin et al. (2024).

The connective's partners are also enriched for well-connected types. 14.4% of connective edges have a rich partner, against $10.2\% \pm 0.1\%$ when each connective node's partners are redrawn uniformly (ratio 1.40, $z = 33.9$, $p = 0.0010$).

The normalized rich-club coefficient of the whole graph exceeds all 100 randomized graphs from total degree 12 upwards, although at that degree the excess is negligible. The coefficient reaches 1.1 at degree 59, 1.5 at 88 and 2 at 172.

Rich-to-rich routing through the connective is modest and fragile

At the 1% threshold, 7,454 two-step routes pass through a connective node from a rich partner on one side to a rich partner on the other. That is 1.103 times the mean of 1000 layer-preserving rewirings ($6,760 \pm 143$, $z = 4.8$, $p = 0.0010$), so the primary hypothesis is supported.

The excess is carried entirely by ascending routes (ratio 1.150, $p = 0.0010$); descending routes are at chance (0.984, $p = 0.71$). The excess also depends on how "rich" is defined. It is significant when the top 5%, 10% or 30% of partners count as rich, and not significant at 1%, 2%, 20% or 50%.

The result also depends on the edge threshold, and the pre-registered robustness hypothesis failed because of it:

The route enrichment grows as weak connections are dropped and disappears when they are added. Rich-to-rich routing through the connective is therefore a property of its strongest connections, not of its wiring as a whole. Placement economy, and the over-representation of connective types among hubs, hold at every threshold.

The connective buys less flow per unit of wire than ordinary wiring, except from brain to nerve cord

The intact graph supports 8,731 edge-disjoint sensory-to-motor paths. Cutting the 36,943 neck-crossing edges (22.37 m of wire) removes 963 of them. Random non-connective sets of the same total length, about 153,000 shorter edges each, remove $2,689 \pm 41$ (ratio 0.36, $z = -42.5$, $p = 1.0$).

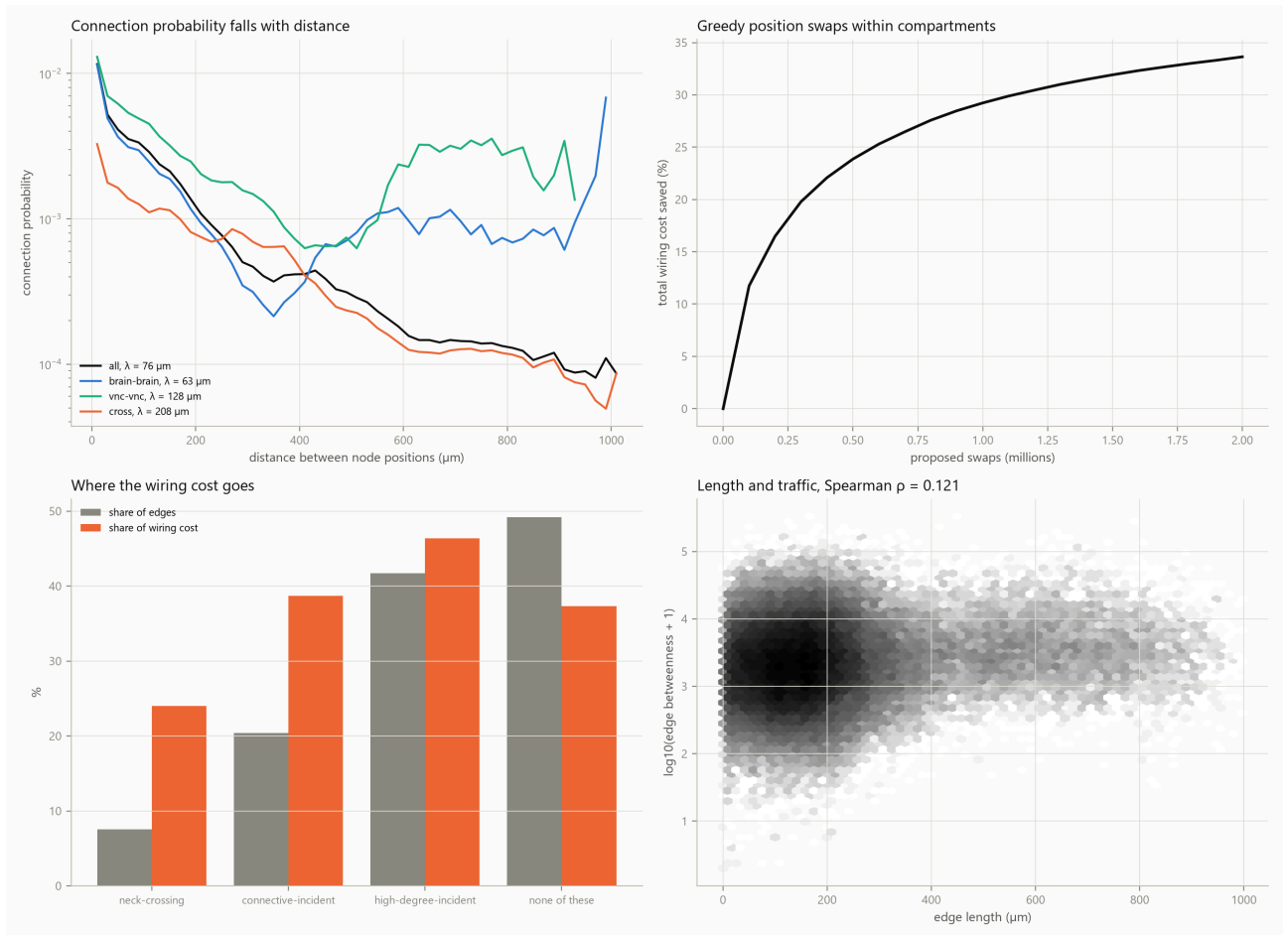


Figure 5: Wiring economy of the cell-type graph: connection probability against distance, cost saved by greedy swaps, the share of edges and of wiring cost by edge group, and edge length against edge betweenness.

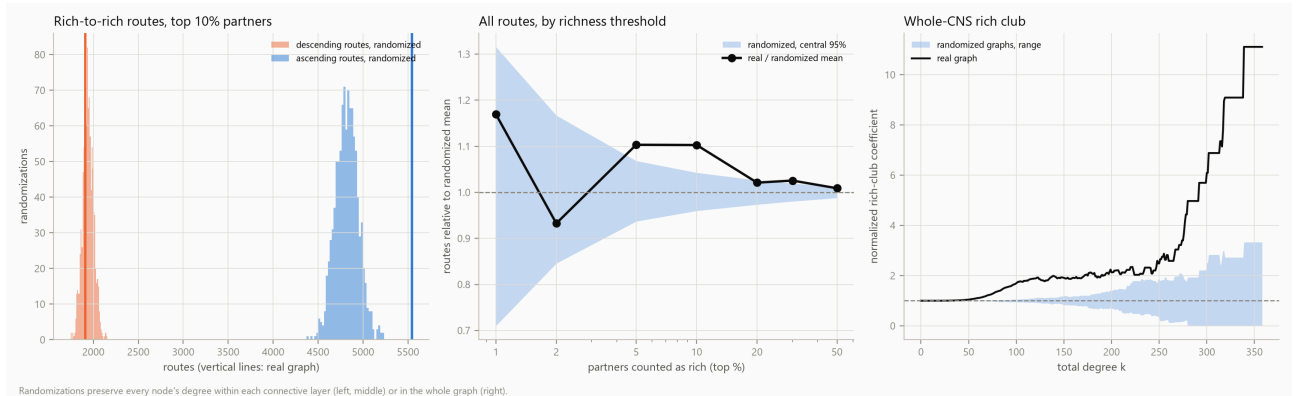


Figure 6: Rich-club structure of the connective. Rich-to-rich routes against layer-preserving rewirings, by richness threshold, and the normalized rich-club coefficient of the whole graph.

Table 5: Robustness to the edge threshold. Placement and hub over-representation hold at every threshold; rich-to-rich routing does not hold at 0.5%.

edge threshold	edges	real / rewired routes	p	hub odds ratio	placement ratio
0.5%	942,400	1.002	0.42	3.99	0.460
1%	490,884	1.103	0.0010	3.22	0.459
2%	216,862	1.391	0.0010	2.22	0.455
5%	52,217	1.605	0.0010	1.33	0.439

Table 6: Value of the neck-crossing edges. Null sets are 1000 random sets of non-connective edges matched on total length (primary) or on edge count; the last row is a single deterministic set.

comparison	flow lost, connective	flow lost, null mean	ratio	p
all flow, equal length (primary)	963	2,689	0.36	1.0
brain sensory → nerve-cord motor, equal length	3,759 of 6,423	2,052	1.83	0.0010
nerve-cord sensory → brain motor, equal length	242 of 2,228	555	0.44	1.0
all flow, equal edge count	963	648	1.49	0.0010
brain → nerve cord, equal edge count	3,759	464	8.10	0.0010
all flow, longest non-connective edges	963	962	1.00	—

The primary value hypothesis is not supported: per unit of wire, neck-crossing wiring carries less sensory-to-motor flow than ordinary wiring. No sensory-motor pair is disconnected by the cut. Every pair the connective joins is also joined by a path that avoids the neck-crossing edges.

The loss is concentrated in one direction:

Cutting the neck removes 59% of the flow from brain sensory to nerve-cord motor nodes, 1.83 times what random wiring of equal length removes. In the other direction it removes 11%, 0.44 times the random loss.

Per connection, neck-crossing edges carry more flow: they remove 1.49 times the flow of an equal number of random non-connective edges.

The comparison that most directly answers whether the connective is special is the last row. The 84,008 longest non-connective edges, each at least 207 μm long, together match the connective's total length. Removing them costs 962 units of flow, one fewer than the connective. On this measure the connective is worth what long wiring of its length is worth. The exception is its brain-to-nerve-cord direction, where the longest edges remove only 1,122 units against its 3,759.

Silencing every edge of the descending and ascending neurons, not only the neck-crossing ones, removes 2,118 units of flow, 0.49 times a length-matched null. It also removes 6,202 of the 6,423 units from brain to nerve cord and 1,543 of the 2,228 from nerve cord to brain, 1.85 and 1.72 times the null. What crosses between brain and nerve cord depends almost entirely on these neurons, the bottleneck Berg et al. (2026) described.

Across cell types, the price of the connective does not predict its value

For a single descending or ascending cell type, removing its neck-crossing edges usually costs no flow at all: this holds for 622 of 911 descending nodes and 899 of 1,010 ascending nodes. The remaining wiring supports the same maximum flow.

Among descending nodes, price and value are correlated ($\rho = 0.231$, $p = 7.6 \times 10^{-13}$). Controlling for the number of neck-crossing edges leaves almost nothing (partial $\rho = 0.044$, $p = 0.09$): at a given number of connections, longer wiring buys no more flow. Among ascending nodes, price and value are unrelated ($\rho = -0.005$). Both price-value hypotheses therefore fail.

The extremes show why:

- The most expensive ascending type, AN07B004, has 316 neck-crossing edges and 146 mm of wire on one side and carries 2 units of flow.
- The descending type DNge002 carries 14 units with 5.5 mm of wire.
- The most expensive descending type, DNge98, is also among the most valuable (123 mm, 16 units).

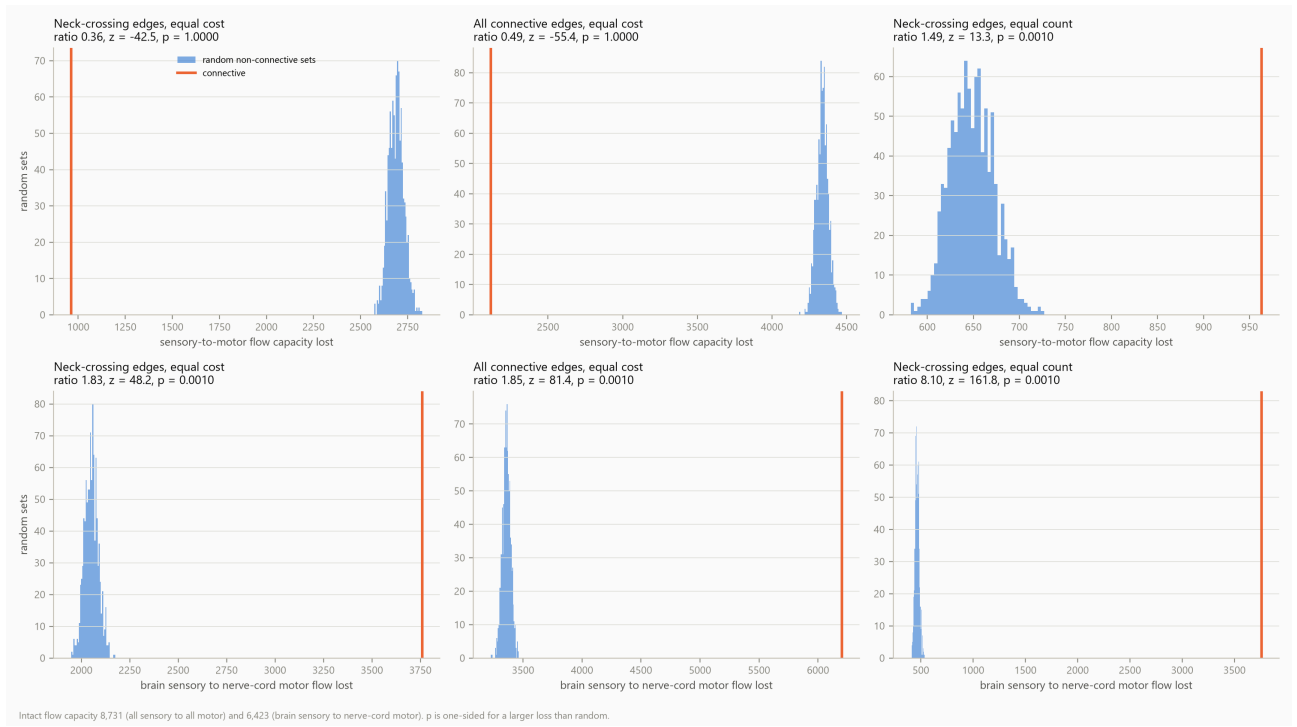


Figure 7: Sensory-to-motor flow lost when the neck-crossing edges are cut, against 1000 cost-matched and count-matched sets of non-connective edges.

A distance and cell-class model reproduces the rich-to-rich routes but little else

Distance alone explains connections moderately well, with McFadden pseudo- R^2 0.165 and cross-validated AUC 0.786. The other terms add fit as follows:

In model G, the log-odds of a connection fall by 0.248 per 100 μm of distance, and a shared compartment adds 0.698. The four largest class-pairing terms all involve descending neurons.

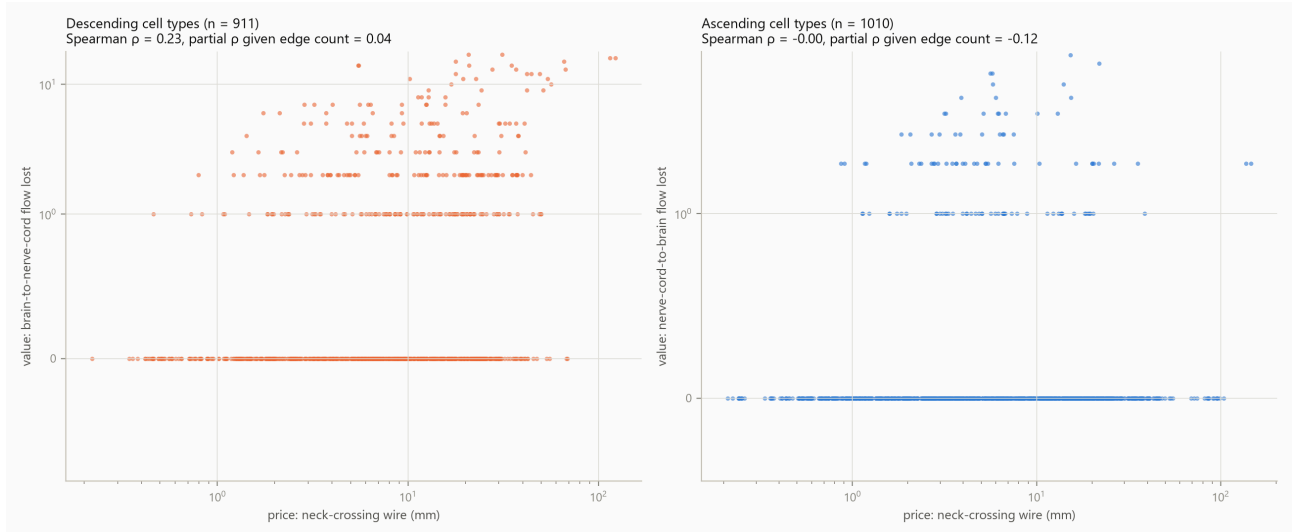


Figure 8: Price and flow value of individual descending and ascending cell types.

Table 7: Fit of the nested generative models.

	model	pseudo- R^2	cross-validated AUC
	distance only	0.165	0.786
1.	compartment	0.165	0.787
1.	superclass pairing (model G)	0.276	0.847
1.	source and target degree (model G+deg)	0.381	0.899

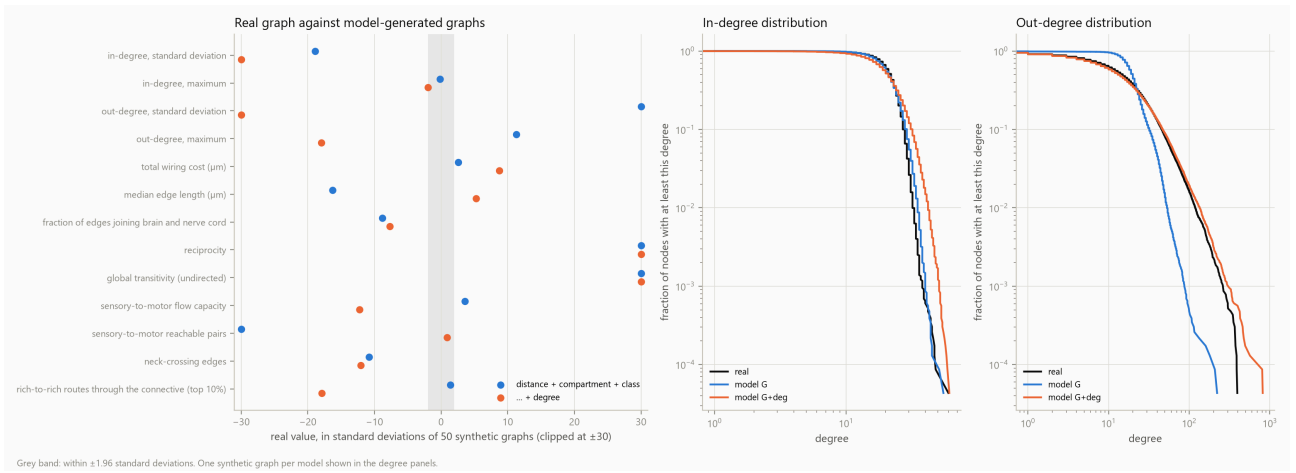


Figure 9: Thirteen properties of the real graph against 50 synthetic graphs from each generative model.

Model G reproduces 2 of the 13 compared properties: the maximum in-degree and the number of rich-to-rich routes through the connective. Its synthetic graphs have 6,636 such routes (central 95% 5,540–7,600) against 7,454 real. The model knows nothing of the real degree sequence. The route count that exceeds degree-preserving rewiring by 10% therefore lies within the range that distance, compartment and cell class produce on their own.

Model G+deg matches the out-degree distribution far better (Kolmogorov-Smirnov distance 0.039 against 0.351). It reproduces only reachable pairs, however, and makes rich-to-rich routes 2.1 times too frequent.

Neither model reproduces the local structure of the real graph, which has 21 and 16 times the reciprocity of their graphs and 16 and 8 times their transitivity, as expected when every ordered pair is drawn independently. Neither reproduces total wiring cost, median edge length, the number of neck-crossing edges (39,241 under model G against 36,943) or flow capacity (8,365 against 8,731).

Discussion

The fly’s CNS is laid out economically in the weaker sense later work settled on, not in the strong sense of Cherniak’s near-optimal layouts. At cell-type resolution, across brain, nerve cord and connective, the real layout is less than half as costly as random placements of the same positions. It is a quarter cheaper even when types are shuffled only within their compartment. The hemibrain neuron shuffles of Salova and Kovács (2025) found real layouts about half as costly as shuffled ones; the ratio here, 0.459, is similar at a different resolution and over the whole CNS.

Like the *C. elegans* layout, the fly’s is far from the cheapest possible arrangement: greedy swaps remove at least a third of its wire, a margin close to Gushchin and Tang’s for worm interneurons. Wiring cost is one pressure among several. The long connections that keep the layout from its optimum are the kind Kaiser and Hilgetag (2006) argued shorten processing paths.

The economy is also local. Attributed to the neuropils, the budget is dominated by a few regions, and inside every region large enough to test the cell types are placed more cheaply than permutations of their own positions. The clearest exceptions are the antennal lobes, whose internal wiring is within a few per cent of the permuted mean in both hemispheres, so whatever fixes the position of a cell type inside an antennal lobe, it is not the length of the connections between them.

How the budget is spread over connections sets the scale for what follows. It is uneven, with a Gini of 0.404 and nearly a third of the wire in the longest tenth of the connections, but the tail is not extreme: a lognormal describes it better than a power law does, and the longest connection spans about the longest axis the somata occupy. Nothing in the length distribution alone marks out a small set of connections as a class apart. What marks the neck-crossing connections out is where they run, not how unusual their length is.

The connective is where those long connections concentrate. It is 7.5% of connections and a quarter of the wire, and in two respects it looks like the hub backbone of mammalian connectomes. Its cell types are over-represented among the most connected types of the CNS, as Lin et al. (2024) predicted, and it favours well-connected partners on both sides.

The result the rich-club framing most directly predicts is that the connective preferentially joins hubs of the brain to hubs of the nerve cord. That result is weaker:

- the excess is 10%, and comes entirely from ascending routes;

- it disappears when weaker connections are counted;
- a model with no knowledge of degree produces as many such routes.

Partner enrichment and hub membership are robust; rich-to-rich routing specifically is not.

The value result reverses the simplest version of the hypothesis. The pre-registered comparison asked whether the connective buys more flow than random wiring of the same length. It buys less, because the same length buys four times as many short connections elsewhere, and flow capacity counts disjoint paths.

Per connection the connective carries more flow than ordinary connections, and against the longest ordinary connections of the same total length it carries the same flow: on this measure it is valued as long wiring in general is valued. Its distinctive contribution is directional, 1.83 times the brain-to-nerve-cord flow of random wiring of its length and less than half of the reverse. Nor does spending more wire buy more flow for individual cell types: most can be removed without loss, and among those that matter, value follows the number of connections rather than their length.

Returning to the question in the title: the fly places its most expensive wiring on well-connected cell types, as the human connectome does. On these measures, however, that wiring does not return more sensory-to-motor capacity per unit length than ordinary long wiring, except in the brain-to-nerve-cord direction.

What this does and does not show

This analysis shows four things about one male fly CNS:

- the placement of its cell types is cheaper than random placement;
- the placement is economical inside the neuropils as well as across them, in every region large enough to test but the antennal lobes;
- the connective's cell types sit among its hubs;
- how much the connective's wiring contributes to one measure of sensory-to-motor routing, measured against three explicitly constructed comparison sets.

It does not show that wiring economy is a law of nervous-system organization, or that the placement is optimal. The generative model reproduces or fails to reproduce specific statistics; it does not describe how the nervous system develops. Flow capacity is a structural count of disjoint paths, not a measure of information transmitted, of behaviour, or of anything cognitive. The results extend the FlyWire brain analyses cited above to data those studies did not have; they do not revise them.

Limitations

- **Resolution.** The graph is aggregated to cell types by hemisphere, and each type is placed at the centroid of its somata. Within a class, soma-to-output distance does not predict a neuron's cable length. Positions therefore capture differences in wiring cost between classes, not the cable of individual neurons.
- **Cost measure.** Euclidean distance between centroids stands in for the length of real processes, which follow neuropil tracts rather than straight lines.
- **Neuropil assignment.** A cell type is attributed to the neuropil surface nearest its position, which is the centroid of its cell bodies rather than of its synapses. Cell bodies lie in a rind outside the neuropils, so the attribution follows where a type's somata sit, and the wire it charges to a region is a property of its endpoints, not of the tract its processes take.
- **Tail fit.** Connection length is bounded by the size of the animal, so the power-law comparison weighs imperfect descriptions of a truncated distribution against each other rather than testing a mechanism.
- **Edge threshold.** Every result uses a 1% input threshold. The placement and hub results hold from 0.5% to 5%; the rich-to-rich route result does not hold at 0.5%.
- **The connective set.** It is defined by majority superclass. Sensory ascending neurons also pass through the neck, but they are treated as sensory inputs because their cell bodies lie outside the CNS. Their neck-crossing wiring is therefore not part of the removed set.
- **Value measure.** Flow capacity counts unweighted edge-disjoint paths, so it rewards many short connections over fewer long ones. For that reason the length-matched comparison is reported alongside count-matched and longest-edge comparisons, not replaced by them.
- **Swap search.** The search is greedy and was stopped before convergence. It also ignores the physical constraints that fix where neuropils and cell-body layers can lie.
- **Generative models.** Both draw each ordered pair independently, so they cannot produce reciprocity or clustering.
- **Data.** The data are one animal, one sex and one static reconstruction.

- **Source version.** The flow analysis of Berg et al. (2026) was read in its bioRxiv version (v2, October 2025); the published version could differ.

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